STATISTICAL DATA ANALYSIS

Poster Presentation Content – Questions 01 to 10

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QUESTION 01

University Learning Management System (LMS) Usage Analysis

Scenario: Most students use the LMS for 2–4 hours weekly; a small number spend over 20 hours. The university wants to understand engagement and academic performance.

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1. Data Summarization

- Record weekly LMS hours, attendance, assignment completion, quiz scores, and final marks.
- Report sample size, mean, median, minimum, maximum, range, quartiles, IQR, and standard deviation. The median is useful when a few students have very high usage.

2. Distribution of Engagement

- Use a histogram to show common usage ranges and a box plot to show median, spread, and potential outliers.
- A scatter plot of weekly LMS hours against marks can show possible associations.

3. Outlier Identification

- IQR = Q3 − Q1. Potential outlier fences are Q1 − 1.5 × IQR and Q3 + 1.5 × IQR.
- Check high hours against login logs, session length, background tabs, and course requirements before treating them as errors.

4. Are Highly Active Students Outliers?

- Not automatically. High usage may reflect strong engagement, a demanding course, accessibility needs, or difficulty understanding material.
- Keep valid observations and interpret them with context rather than labeling students negatively.

5. Skewness and Kurtosis

- A long right tail suggests positive skewness because most students use the LMS for a few hours and a few use it much longer.
- Kurtosis describes tail heaviness and extreme values. Calculate both from real records; the scenario alone does not provide numerical values.

6. LMS Usage and Performance

- Use Pearson or Spearman correlation and compare marks across usage groups.
- Consider prior achievement, course difficulty, attendance, and assignment completion. Correlation does not prove that LMS time causes better marks.

7. Recommendations

- Support students with low activity or missing assignments; offer study-skills help to high-use students with low marks.
- Improve navigation, mobile access, reminders, and instructor feedback while protecting student privacy.

Suggested Poster Visuals

- Histogram of weekly LMS hours
- Box plot of LMS hours
- Scatter plot of LMS hours versus marks

Conclusion

Use LMS activity with assessment results and feedback. High usage can indicate engagement or difficulty; investigate context.

QUESTION 02

Renewable Energy Production Analysis

Scenario: Solar production is usually 5,000–8,000 kWh daily, exceeds 12,000 kWh on exceptionally sunny days, and may fall below 2,000 kWh in cloudy or rainy weather.

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1. Descriptive Statistics

- Calculate n, mean, median, minimum, maximum, range, quartiles, IQR, standard deviation, and coefficient of variation.
- Check missing readings, units, equipment downtime, and sensor calibration.

2. Distribution

- Use a histogram, box plot, and daily time-series chart to examine spread, seasonal patterns, and multiple peaks.
- Compare distributions by clear, partly cloudy, cloudy, and rainy conditions.

3. High and Low Values

- Flag values outside IQR fences or engineering limits for review, not automatic removal.
- Low production may result from weather, shading, maintenance, inverter faults, or sensor errors; high production may be genuine or caused by a reporting error.

4. Handling Outliers

- Verify timestamps, units, sensor calibration, and plant status.
- Correct confirmed errors with a documented reason. Keep genuine weather extremes and consider robust statistics or event indicators.

5. Skewness and Kurtosis

- Skewness describes whether the distribution has a longer high or low tail. Kurtosis describes tail heaviness.
- Calculate both by season or weather group and state the kurtosis definition used.

6. Weather Impact

- Compare generation by weather category and include irradiance, temperature, cloud cover, rainfall, and daylight hours where available.

7. Forecasting and Operations

- Compare a baseline time-series forecast with a weather-informed model; validate on later dates and report MAE or RMSE.
- Schedule maintenance during low-generation periods and plan storage or grid supply for forecast shortfalls.

Suggested Poster Visuals

- Histogram of daily kWh
- Box plot of production

- Time series with weather categories

Conclusion
Weather-aware forecasts can improve scheduling, storage, and energy planning. Keep genuine weather extremes and correct verified errors.

QUESTION 03

E-Commerce Festival Sales Analysis

Scenario: Average daily sales are ₹10 lakh in normal periods and rise to ₹2 crore per day during a festival season.

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1. Summary Statistics

- Calculate mean, median, minimum, maximum, quartiles, IQR, standard deviation, order count, and average order value separately for normal and festival periods.
- ₹2 crore = ₹200 lakh, which is 20 times the stated normal average of ₹10 lakh.

2. Distribution Changes

- Festival demand may shift sales upward and increase spread. Use side-by-side histograms, box plots, and a time-series chart.

3. Unusual Patterns

- Check spikes, sudden drops, duplicate transactions, refunds, bot orders, promotions, order counts, website traffic, and payment records.

4. Are Festival Sales Outliers?

- Not automatically. Festival sales are expected event-driven observations. Keep valid values and label the event period; model normal and festival demand separately if useful.

5. Skewness and Kurtosis

- Sales are often right-skewed because a few days or products contribute very high revenue. High kurtosis may indicate heavy tails or extreme sales days.
- Calculate these measures separately for normal and festival periods.

6. Business Insights

- Identify top-selling products, peak dates, customer segments, conversion rates, promotion effectiveness, profit margins, and returns.

7. Inventory Recommendations

- Forecast product-level demand using past festivals, promotions, supplier lead times, and capacity.
- Set safety stock and reorder points; coordinate staffing, delivery, customer support, and returns processing.

Suggested Poster Visuals

- Side-by-side sales bar chart
- Box plots before and during festival
- Daily sales time series

Conclusion

Festival sales are meaningful event-driven data. Forecast event demand separately to plan stock, staffing, and delivery.

QUESTION 04

Healthcare Monitoring System

Scenario: Most wearable heart-rate readings are 60–90 bpm, while some readings exceed 180 bpm.

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1. Data Summarization

- Summarize heart rate by patient and time window using mean, median, minimum, maximum, percentiles, variability, missingness, and duration outside configured limits.

2. Visualization

- Use time-series charts, distribution plots, and dashboards with timestamps, alert thresholds, and sensor-quality indicators.

3. Abnormal Readings

- Use clinically configured thresholds, sudden-change rules, repeated readings, and persistence over time.
- A reading above 180 bpm may be serious, but interpretation depends on age, activity, symptoms, and clinical context.

4. Erroneous Sensor Values

- Check sensor contact, motion artifacts, signal quality, timestamps, and calibration. Mark uncertain readings and document corrections; do not silently delete abnormal values.

5. Skewness and Kurtosis

- Calculate distribution measures by patient or meaningful group. A high-rate tail may cause positive skewness; kurtosis may highlight extreme readings, but neither diagnoses a patient.

6. Emergency or Data Error?

- Follow clinical escalation protocols and repeat measurements when safe. Do not delay urgent assessment for statistical checks; qualified clinicians determine medical significance.

7. Real-Time Recommendations

- Use reliable sensors, patient-specific thresholds, alert escalation, and clear handoffs.
- Protect records with access controls and encryption. Monitor false alarms and missed events; statistical detection never replaces clinical judgment.

Suggested Poster Visuals

- Heart-rate time series with alert limits
- Histogram of valid readings
- Readings versus time

Conclusion

Abnormal heart-rate readings require data-quality checks and clinical context. Do not ignore possible emergencies as statistical outliers.

QUESTION 05

Cybersecurity Network Traffic Analysis

Scenario: Normal traffic is about 20,000 requests per hour, but traffic rises to 2 million requests per hour during a suspected attack.

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1. Traffic Summary

- Measure requests per time interval, source IPs, endpoints, response codes, bandwidth, latency, failed logins, and concurrent connections.
- Report median, percentiles, peak rate, and baseline variability.

2. Normal Distribution

- Build baselines by hour, weekday, endpoint, and season. Plot request-rate time series and histograms.

3. Detecting Spikes

- Use rolling baselines, robust z-scores, percentiles, change-point detection, and service-capacity thresholds.
- The increase from 20,000 to 2,000,000 requests/hour is 100-fold and warrants investigation.

4. Outlier Treatment

- Do not discard attack-period traffic. Label incidents, correct confirmed telemetry errors, preserve raw data, and use robust methods to protect the baseline.

5. Skewness and Kurtosis

- Traffic often has a long right tail and occasional bursts. Use log-scale plots where useful but retain original counts for interpretation.

6. Anomaly Detection

- Combine statistical thresholds with attack signatures, source reputation, request patterns, and threat intelligence. An alert indicates suspicion, not proof of an attack.

7. Security Measures

- Use DDoS protection, web application firewalls, rate limits, bot detection, patching, strong authentication, least privilege, and network segmentation.
- Centralize logs and maintain tested incident-response and backup plans.

Suggested Poster Visuals

- Requests-per-hour time series
- Log-scale traffic histogram
- Traffic spike highlighted on a chart

Conclusion

Traffic spikes may be attacks or legitimate demand. Combine statistical alerts with security telemetry and response procedures.

QUESTION 06

Manufacturing Quality Control

Scenario: A company produces 50,000 bottles daily with a target of 1,000 ml. Some bottles contain less than 900 ml or more than 1,100 ml.

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1. Data Summary

- Record fill volume, line, machine, shift, batch, and time. Calculate mean, median, standard deviation, range, quartiles, IQR, and defect rate.

2. Distribution

- Plot a histogram with the 1,000 ml target and specification limits; compare box plots by line or shift.

3. Detecting Defects

- Use specification limits and process-control methods, not only IQR outlier rules. Apply the official acceptance criteria required for the product.

4. Abnormal Measurements

- Verify scale calibration, sampling, units, and data entry. Re-measure suspect bottles when feasible.
- Keep genuine defects in quality reports; exclude only confirmed measurement errors with documented reasons.

5. Skewness and Kurtosis

- Skewness may reveal a shift toward underfill or overfill. Kurtosis can indicate heavier tails and more extreme volumes. Use these with control charts.

6. Quality Impact

- Underfill can breach label claims and reduce trust; overfill increases material costs. Defects may cause complaints, rework, recalls, and regulatory problems.

7. Process Improvement

- Calibrate filling machines, use control charts, investigate root causes by line and shift, train operators, and verify corrective actions with follow-up samples.

Suggested Poster Visuals

- Fill-volume histogram with 1 L target
- Control chart of sample means
- Box plot by production line or shift

Conclusion

Monitoring can identify underfill, overfill, and process drift. Correct root causes and verify instruments before removing observations.

QUESTION 07

Climate Change and Weather Analysis

Scenario: Temperatures are usually 25°C–35°C, but several days in the last decade exceeded 45°C.

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1. Temperature Summary

- Summarize daily minimum, maximum, and mean temperatures using mean, median, range, quartiles, standard deviation, and percentiles.
- Report by season and year, not only as one overall average.

2. Climate Distribution

- Use histograms, seasonal box plots, and time-series charts. Check missing readings and changes in weather stations.

3. Extreme Events

- Identify days above a pre-defined heat threshold and compare with historical percentiles. Verify station records and nearby readings.

4. Outlier Treatment

- Do not automatically remove extreme temperatures: they may be genuine climate events. Correct confirmed instrument or transcription errors and document uncertain values.

5. Skewness and Kurtosis

- Positive skewness may indicate a long tail of hot days; kurtosis can describe extreme-value frequency. Calculate by season and period.

6. Research Implications

- Analyze long-term trends, heatwave duration, threshold exceedances, and nighttime heat. Attribution to climate change requires suitable methods and broader evidence.

7. Policy Recommendations

- Develop heat alerts, cooling centers, public communication, shade and tree cover, cool roofs, worker protections, and heat-resilient urban design. Maintain quality-controlled weather data.

Suggested Poster Visuals

- Temperature time series
- Histogram of daily maximum temperatures
- Box plots by season or decade

Conclusion

Extreme temperatures are important climate signals. Use quality-controlled long-term data to support heat adaptation.

QUESTION 08

Social Media Engagement Analysis

Scenario: A creator uploads 500 videos. Most receive fewer than 10,000 views, while a few viral videos exceed 10 million views.

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1. Data Summary

- Summarize views, watch time, likes, comments, shares, retention, and engagement rate with median, mean, percentiles, and spread.
- Record upload date, topic, duration, format, and paid promotion.

2. Distribution

- Views are likely strongly right-skewed. Use a histogram with a log-scaled axis, box plot, and scatter plot of views versus watch time or engagement rate.

3. Viral Outliers

- Use IQR fences or robust log-view comparisons to flag videos. Verify genuine views and check promotions, collaborations, and external events.

4. Retain Viral Videos?

- Yes, if valid. Viral videos are important outcomes, not automatically errors. Analyze them separately for typical performance comparisons, but keep them for total reach analysis.

5. Skewness and Kurtosis

- A long right tail gives positive skewness. High kurtosis may reflect a few exceptionally high-view videos. Explain whether calculations use raw or log-transformed views.

6. Business Implications

- Viral content may increase followers and sponsorship value, but views do not guarantee revenue or loyalty. Check retention, shares, conversion, revenue, and audience fit.

7. Engagement Strategies

- Study successful themes, hooks, formats, duration, and posting times. Use comments and retention curves, test one change at a time, and maintain a consistent schedule.

Suggested Poster Visuals

- Log-scale histogram of video views
- Box plot of views
- Views versus watch time

Conclusion

Viral videos can reveal audience preferences. Retain valid viral observations and analyze them separately when useful.

QUESTION 09

Stock Market Volatility Analysis

Scenario: A stock usually changes within ±2% daily but rises by 30% in one day after a major corporate announcement.

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1. Return Summary

- Calculate daily percentage or log returns and summarize mean, median, standard deviation, min/max, percentiles, and rolling volatility. Adjust prices for splits and dividends where relevant.

2. Distribution

- Use return time series, histogram, box plot, and Q-Q plot. Financial returns can have heavy tails and changing volatility.

3. Detecting Abnormal Moves

- Compare the 30% return with historical percentiles, market and sector movements. Verify price data, corporate actions, and the announcement date.

4. Extreme Observations

- Keep verified returns. Use robust estimation, event indicators, or separate event analysis rather than deleting genuine extremes. Correct only confirmed data errors.

5. Skewness

- Positive skewness means a longer or more extreme positive-return tail; negative skewness means a longer negative-return tail. Interpret with sufficient data.

6. Kurtosis and Risk

- High excess kurtosis indicates heavier tails than a normal distribution and more frequent extreme outcomes. This matters for stress testing and risk estimates.

7. Investment Recommendations

- Do not decide from one extreme return alone. Review fundamentals, valuation, liquidity, diversification, concentration, and risk tolerance. Use risk controls and acknowledge the possibility of loss.

Suggested Poster Visuals

- Histogram of daily stock returns
- Return time series with event date
- Box plot or Q-Q plot

Conclusion

Extreme returns affect financial risk. Investigate context and use robust risk measures rather than relying on one unusual day.

QUESTION 10

Smart City Traffic Management System

Scenario: Average traffic is 700 vehicles per hour, increasing to 7,000 vehicles per hour during a major cultural festival.

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1. Traffic Summary

- Summarize counts by road, direction, lane, interval, weekday, and event type. Calculate mean, median, peak, percentiles, variability, and missingness.

2. Distribution

- Use hourly line charts, histograms, and box plots for ordinary days, weekends, and event days. Compare roads and peak hours.

3. Unusual Events

- The rise from 700 to 7,000 vehicles/hour is ten times the stated average. Check sensor calibration, duplicate counts, diversions, and event schedules.

4. Outlier Methods

- Use IQR fences, rolling baselines, seasonal thresholds, and change-point detection. Keep genuine festival traffic and label it; correct confirmed sensor faults only.

5. Skewness and Kurtosis

- Traffic may be right-skewed with heavy tails due to event surges. Calculate measures by road and time period and interpret alongside time-series charts.

6. Management

- Adjust signal timing, deploy traffic officers, offer real-time route guidance, increase public transport, manage parking, protect pedestrians, and coordinate closures with organizers.

7. Predictive Model

- Use historical counts with hour, weekday, season, weather, roadworks, and event calendar features. Start with a seasonal baseline, then compare regression or time-series models.
- Train on earlier dates and validate on later dates; report MAE or RMSE and compare predicted with actual festival counts.

Suggested Poster Visuals

- Hourly traffic line chart
- Box plot by day type
- Predicted versus actual traffic

Conclusion

Festival traffic should be forecast using event calendars, weather, and historical counts to improve transport, signal timing, and safety.

Common Statistical Reference

Useful Formulas

- Range = Maximum − Minimum.
- Interquartile range (IQR) = Q3 − Q1.
- Potential outlier fences: Q1 − 1.5 × IQR and Q3 + 1.5 × IQR.
- Coefficient of variation = Standard deviation ÷ Mean × 100%, when the mean is meaningful and non-zero.
- MAE = average of absolute differences between actual and predicted values.
- RMSE = square root of the average squared difference between actual and predicted values.

Important Interpretation Notes

- Skewness describes asymmetry; kurtosis relates to tail weight and extreme observations. State whether ordinary kurtosis or excess kurtosis is reported.
- Statistical outliers are not automatically errors. Validate observations and use subject-specific context.
- In healthcare and cybersecurity, unusual values may be the most important observations and should not be automatically removed.
- Correlation does not establish causation. Evaluate predictive models on data not used for training.